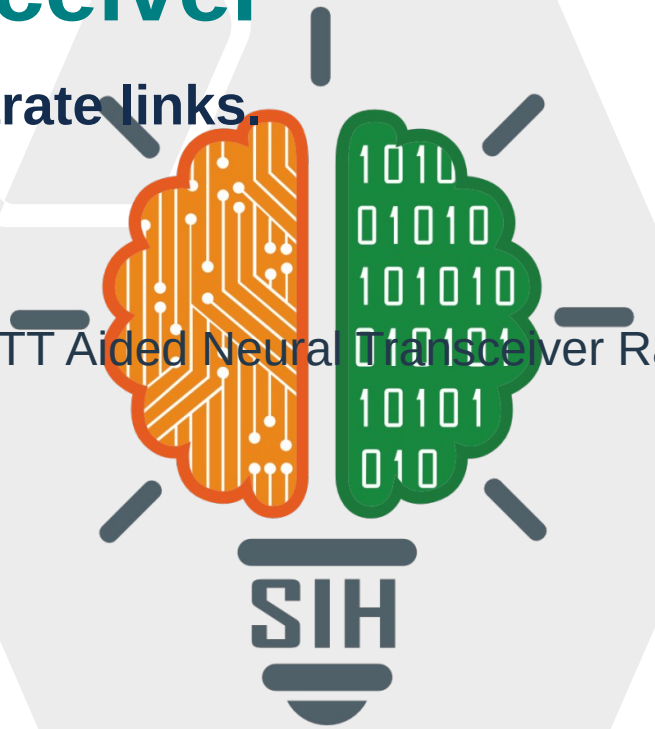
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iTantra — Offline Voice Transceiver

Speech to text to speech, fully offline, over low-bitrate links.



Problem Statement ID – SIH26173

Problem Statement Title – iTantra - Indian Multilingual TTS & STT Aided Neural Transceiver Radio Access for low bitrate links

Theme – Smart Automation

PS Category – Software

Team ID – [PORTAL TEAM ID]

Team Name (Registered on portal) – B.J.P (Bug Janta Party)

PROPOSAL WITH LOCAL MEASUREMENTS • Integrated two-phone prototype and Android performance not yet validated

IDEA TITLE

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iTantra | send voice as text, not as audio

Proposed Solution (Describe your Idea/Solution/Prototype)

An offline Android app: speech in, text out, tiny packet over Wi-Fi/Bluetooth, speech back out on the receiving phone.

Detailed explanation of the proposed solution

Push-to-talk works like a walkie-talkie. When idle, both phones run a full-duplex phone-style path. Alerts play at maximum volume and cannot be interrupted.

How it addresses the problem

Audio is too large for low-data-rate links. One spoken sentence becomes ~50-110 bytes of UTF-8 text — roughly 1,000x smaller — so voice fits where it could not before.

Innovation and uniqueness of the solution

Asymmetric pipeline: the speaker's phone carries the ASR cost, the listener's phone carries TTS. Idle listening runs only a voice-activity detector, so idle CPU stays near zero by design.

TECHNICAL APPROACH

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Technologies to be used (e.g. programming languages, frameworks, hardware)

Kotlin Android app • sherpa-onnx (open-source, ONNX runtime) • Silero VAD 629 KB, RTF 0.004

ASR: Meta Omnilingual CTC 300M INT8, Apache 2.0 • TTS: Piper VITS voices, MIT licence • Wi-Fi Direct / Bluetooth RFCOMM transport

Methodology and process for implementation (Flow Charts/Images/ working prototype)

EITHER PHONE — TALKING

Mic 16 kHz mono

Silero VAD cuts sentence

Omnilingual ASR → text

Language filter → packet

→

TRANSPORT

Bluetooth RFCOMM (primary)

Wi-Fi Direct fallback

UTF-8 text only

~50–250 bytes per message

→

EITHER PHONE — RECEIVING

Piper TTS (hi / en)

Audio out 22050 Hz

Max volume, no interrupt

Alert loops until read

Desktop: ASR RTF 0.039–0.043, TTS RTF 0.032–0.044. On Android: ASR RTF 0.120–0.127, TTS RTF 0.055–0.058, 536 MB PSS with all three models resident. Mouth-to-ear ≈ 0.9–1.0 s.

All Android figures measured on an API 34 arm64 emulator, not a low-end phone. A real budget handset will be slower.

FEASIBILITY AND VIABILITY

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Analysis of the feasibility of the idea

- Every component is open-source and permissively licensed (Apache 2.0 / MIT).
- All 10 required languages scored locally on real human speech.
- An official Android reference app already runs this exact ASR model with VAD.

Potential challenges and risks

- Latency and RAM measured on an emulator, not a low-end phone.
- Under noise, Hindi script retention falls to ~50% at 15 dB SNR.
- TTS voices: 7 of 10 neural; Kannada, Tamil, Odia use espeak-ng.
- Wi-Fi Direct behaves differently across Android vendors.

Strategies for overcoming these challenges

- Benchmark on the cheapest available phone first.
- Report CER alongside WER for complex scripts.
- Scope the demonstrator to Hindi + English, then extend.
- Ship Bluetooth fallback; never depend on one transport.

Measured accuracy — all 10 required languages, 200 real-speech clips each (FLEURS)

Normalised character error rate — lower is better

Kannada 0.053 · Malayalam 0.059 · English 0.067 · Gujarati 0.071 · Bengali 0.100

Telugu 0.100 · Marathi 0.108 · Hindi 0.113 · Tamil 0.129 · Odia 0.134

Every required language measured; all under 0.14. TTS round-trip: English 0.030 (strong), Hindi 0.339 (weak — open problem).

IMPACT AND BENEFITS

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Potential impact on the target audience

For field responders, volunteers and citizens on degraded or saturated networks: spoken alerts and instructions that reach the other end as intelligible speech, without requiring literacy or an internet connection.

Benefits of the solution (social, economic, environmental, etc.)

SOCIAL / proposed

Voice in, voice out — inclusive of users who cannot read or type. Alerts are heard, not read.

ECONOMIC / proposed

No paid API, no cloud, no proprietary SDK. Runs on phones people already own; only free open-source models.

ENVIRONMENTAL / proposed

Text instead of audio lowers transmitted energy per message; effect on battery runtime must still be measured.

Measured so far: all 10 required languages scored on 200 real-speech clips each; Hindi and English round-trip with meaning intact; ~1,000x bandwidth reduction per sentence.

Not yet claimed: field accuracy in noise, phone-class latency, human TTS legibility panel, battery runtime.

Benefits are proposed outcomes, not measured results. No claim of lives saved or guaranteed delivery at this stage.

RESEARCH AND REFERENCES

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Details / Links of the reference and research work

Models and runtime

- csukuangfj/sherpa-onnx-omnilingual-asr-1600-languages-300M-ctc-int8-2025-11-12 — ASR, Apache 2.0, 348 MiB
- rhasspy/piper-voices hi_IN pratham and en_US lessac — TTS, MIT licence, ~60 MiB each
- k2-fsa/sherpa-onnx — Silero VAD (629 KB) and the SherpaOnnxVadAsr Android reference app
- Pre-built APK verified: sherpa-onnx-1.13.8-arm64-v8a-vad_asr-1600-omnilingual_asr_300M_ctc_int8.apk, 284 MiB

Evaluation data

- google/fleurs (CC-BY-4.0) — real recorded human speech used for per-language accuracy

Platform documentation

- k2-fsa.github.io/sherpa/onnx — VAD, ASR, TTS and Android guides
- developer.android.com — Wi-Fi Direct and Bluetooth P2P transport

Local evidence: fleurs_eval_full.py, tts_intelligibility.py, measure_ram_all.py, end_to_end.py — all reproducible from this submission.